

System and method for analog matrix computation using inverted meta-learning.

Technical Field

The invention relates to the field of computing hardware for artificial intelligence, more specifically to a hybrid architecture for analog in-memory computing ("In-Memory Computing"). The invention combines two analog correction matrices and one analog primary computation matrix with a digital control unit to form a joint hybrid atomic triad. These physical substrates, consisting of multiple hybrid atomic matrix triads, execute meta-learning algorithms to achieve digital determinism in a non-deterministic analog medium.

Background of the Invention

Conventional digital systems for neural networks are strictly deterministic but suffer from high power consumption due to the so-called "memory wall" (data transfer between memory and processor). Analog computing systems offer high energy efficiency but are limited by physical non-idealities such as thermal drift, voltage leakage, and component variations.

Existing meta-learning (such as "MAML" Model-Agnostic Meta-Learning) is primarily used to adapt digital models to new external tasks (referred to as "few-shot learning"). There is currently a lack of systems that use intrinsic ("inward") meta-learning to compensate for internal variations in analog computing hardware in real-time during operation.

One of the most limiting factors in today's digital AI systems is the so-called Von Neumann bottleneck, often referred to as the 'memory wall'. In matrix-vector multiplication, which constitutes the majority of computations in an LLM, large amounts of data (weights) must constantly be moved between the memory and the processor. The energy consumption for moving this data is often several orders of magnitude higher than for the computation itself. This makes large-scale AI energy-intensive and difficult to implement in resource-constrained edge systems.

A pervasive problem for today's leading actors in analog in-memory computing (IMC)—whether based on resistive elements (e.g., memristors) or the charge domain (e.g., switched capacitors)—is the difficulty of maintaining deterministic precision over time. To make analog cell intersections function consistently, manufacturers are forced to rely on tight, complex, and costly manufacturing tolerances in the fabrication process for ASIC circuits. Furthermore, current IMC technology relies primarily on static, pre-calibration (offline calibration) and hardware-aware training (Hardware-Aware Training). This means attempting to fix the system so that the analog cells remain deterministic throughout their entire operating lifetime. However, this static approach fails to handle dynamic fluctuations that occur during operation, such as thermal drift, temporary voltage drops, or long-term component aging. Thus, there is a significant and unfulfilled need for an architecture that breaks free from the requirement of perfect physical manufacturing precision and static determinism. There is a lack of a solution that instead allows the analog hardware to be naturally imperfect, but which can dynamically and autonomously adjust and compensate for these internal variations in real-time, without losing either the energy efficiency of the analog computation or the reliability of digital systems.

Summary of the Invention

The core of the invention is an inverted MAML architecture. In conventional MAML, a model is adapted to a changing environment, but here, a digital reference model (a deterministic truth) is used instead to adapt an analog computing core—a so-called "atomic triad" (3, 8, 33) representing a non-deterministic reality. The analog correction matrices (3, 8) and the activation functions (4, 10) together constitute the inverted MAML structure. The system thereby learns to continuously adjust, during operation, the errors in the third analog matrix (33) using short windows for correction measurements.

Through this architecture, a form of technical proprioception is introduced, wherein the atomic computing cores are allowed to become aware of their own local "body" and its physical state. In this context, proprioception is defined as the system's ability to register and compensate for internal physical parameters, such as thermal drift and voltage leakage, in real-time via the reference nodes (29) and the meta-learning algorithm, without external calibration data.

By leveraging the MAML algorithm's capacity for rapid adaptation via few measurement points (few-shot learning), each computing core is arranged to autonomously perform calibration during short temporal windows (20, 24). This internal correction cycle operates at a lower frequency than the primary computing path (payload), allowing local hardware deviations to be compensated for in real-time without affecting the system's throughput or latency during the inference phase.

Through an atomic triad structure, complex non-linearities are eliminated by letting the primary matrix (33) perform the main computation while the cascaded layers, the correction matrices (3, 8), add an inverted MAML deviation correction at the summing point (35). The resulting output signal (11) is controlled via feedback (13) to deterministically correspond to the digital reference model (39).

To maintain precision in gradient extraction and avoid temporal blur (integration losses) during the discharge phase (31), short temporal windows (20, 24) are used. Instead of long measurement periods, the measured values are downsampled by placing snapshot windows at exactly the same phase position across multiple refresh cycles (18). This enables high instantaneous precision even for large matrices without the information being averaged over time within the measurement window.

The system is trained into a robust base model (40, 41) through an emulation process corresponding to a changing hardware state during operational inference (42). Instead of physical relocation of different substrates, emulated operating variations for the power supply of the analog matrices are injected via a controllable DAC (trim-DAC) (38). During base model training (40, 41), small unknown offset values are also added for the MAML training (40, 41) to weights, the DAC (2), and the ADCs (45, 49) to emulate the physical deviations of the analog medium. By varying these emulated errors during training, the base model is exposed to the total statistical variations of the hardware population. The inverted MAML model thus becomes proficient at rapidly re-trimming itself against variations during operation. Upon transition to normal operational inference drift (42), all emulated offset values are removed and said trim-DAC (38) is reset. This creates a base model that, with a minimal number of samples and via an inner calibration loop (41), can be fine-tuned to the natural state of the specific physical core, even during ongoing inference (19) to the system's output signal (11).

This adaptability is enabled by the deviation correction in the components (3, 4, 8, 10) being specifically designed as a multi-layer non-linear architecture. The purpose is to streamline the system's modeling capability to exclusively map and invert underlying physical hardware errors, rather than burdening the meta-learning algorithm with learning the primary logical computation, which remains fixed in the primary matrix (33). By separating the primary inference computation (payload computation)—whose applied matrix weights are deterministic and fixed, but whose analog output signal inevitably varies—from the adaptive non-linear error correction, sufficiently accurate compensation is made possible (for example, 6-bit determinism on the outgoing vector computation compared to the computation of the ideal digital model). When the output signal of the error correction is linearly summed with the fluctuating output from the primary matrix at the summing point (35), the computational deviations are neutralized, whereby the system achieves digital precision despite the inherent variations of the analog medium.

By utilizing the proposed hybrid architecture (Fig. 1), the need to move weights during the inference phase itself (19) is eliminated, as the weights are physically stored in the analog matrix (3, 8, 33). A crucial aspect of the invention is the asymmetry in memory bandwidth between computation and control (Fig. 1).

While the analog matrices (3, 8, 33) perform numerous parallel computations with high throughput, the digital control unit (9) operates with a memory bandwidth that is orders of magnitude lower. This is possible because the digital control unit (9) only needs to communicate the fine-tuned correction values (via the MAML algorithm) in a slow loop (12, 13, 14, 15, 16, 17) to maintain sufficiently high determinism. By separating the high-frequency data throughput (analog) from the low-frequency calibration (digital), a significant reduction in total system power is obtained without losing precision.

An important decisive technical advantage of the invention is the functional synergy between the rapid convergence of the MAML algorithm and the analog hardware architecture's need for calibration during operation. The system is arranged to operate with an optimized duty cycle (Fig. 5), where the majority of the operation is dedicated to uninterrupted inference (19), while a minimal fraction of the time—preferably on the order of a few percent—is reserved for automatic fine-calibration. Since the inverted MAML algorithm is specifically trained to achieve adaptive adjustment with a minimal number of snapshot measurement readings (samples) (referred to as 'few-shot learning'), a complete fine-tuning of the analog weights is made possible within the short time windows (20, 24) available between the matrix's refresh pulses (18). This symbiosis between algorithm and hardware architecture means that the system can maintain maximum data throughput while digital determinism is ensured through rapid, periodic correction. This ability to interpolate and transition between different analog states with few measurement points makes the combination of MAML and the proposed hybrid matrix structure (Fig. 1, 2, 3, 4) uniquely suited for demanding real-time AI applications.

As an integrated part of this resource efficiency optimization, the system applies a time-distributed acquisition strategy for gradient extraction. Instead of requiring an extensive and energy-consuming measurement window in each individual cycle, the collection of measurement data from the specific points in time (21, 22, 23) is spread out over multiple subsequent refresh cycles (18).

Through this downsampled method, the digital control unit (9) can successively build up a complete picture of the state of the matrix by sampling different parts of the weight matrix at exactly the same relative points in time during the process, but in different chronological cycles. This allows for an asymmetric update where large amounts of analog weights can be fine-tuned with a minimal digital bandwidth. This capability to perform high-precision calibration via a distributed measurement series ensures that the system maintains its determinism over time, without encroaching on the real-time performance required for continuous inference (19).

To further optimize this calibration process and maximize the signal-to-noise ratio (SNR), the system implements a parallel encoding unit (6) (excitation device) based on orthogonal encoding, preferably Hadamard encoding (43) (Fig. 9). By exciting all rows in the matrix simultaneously with unique, orthogonal patterns (Fig. 9), the digital control unit (9) can isolate individual cell deviations via a Fast Hadamard Transform (FHT). In contrast to a serial sequence, where each row is analyzed in isolation (conventional sequential row scanning), this parallel method allows the perturbation energy to be distributed across all rows during the measurement cycle. This results in a significant lowering of the noise floor and thus a higher precision in the estimated gradients, without requiring an extension of the temporal measurement window's duration. The method maintains the system's critical charge neutrality across the weight capacitors (44), while adding a statistical robustness that is necessary to ensure determinism in large-scale matrices where thermal noise would otherwise limit calibration accuracy.

To ensure maximum correlation between reference measurements and actual computational processes, the matrix is arranged such that the dedicated reference cells (29) (according to claim 2) are physically interleaved with the active computing cells (27) for both the first (3) and second (8) layers. This complete interdigitation (30) means that the reference cells (29) are exposed to identical local thermal fluctuations and process variations as the active cells, which eliminates measurement errors caused by spatial distances on the semiconductor substrate.

Through the proposed hybrid architecture (Fig. 1), the need for complex and time-consuming simulations of the analog hardware is eliminated. Since the system, via the digital control unit (9), adapts to the actual physical variations in real-time, the requirement to create precise deterministic models of the analog medium is bypassed.

Figurförteckning

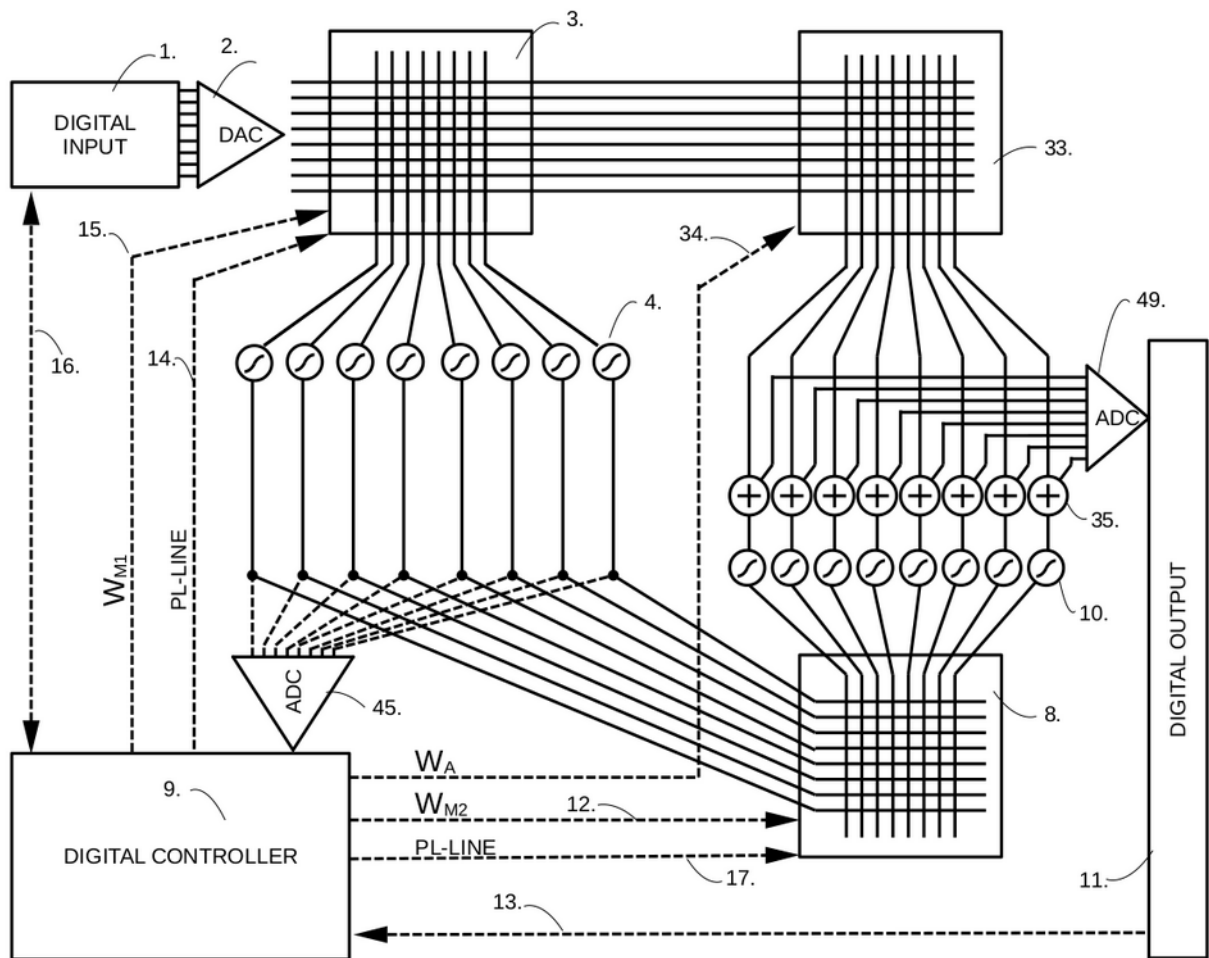


Fig 1

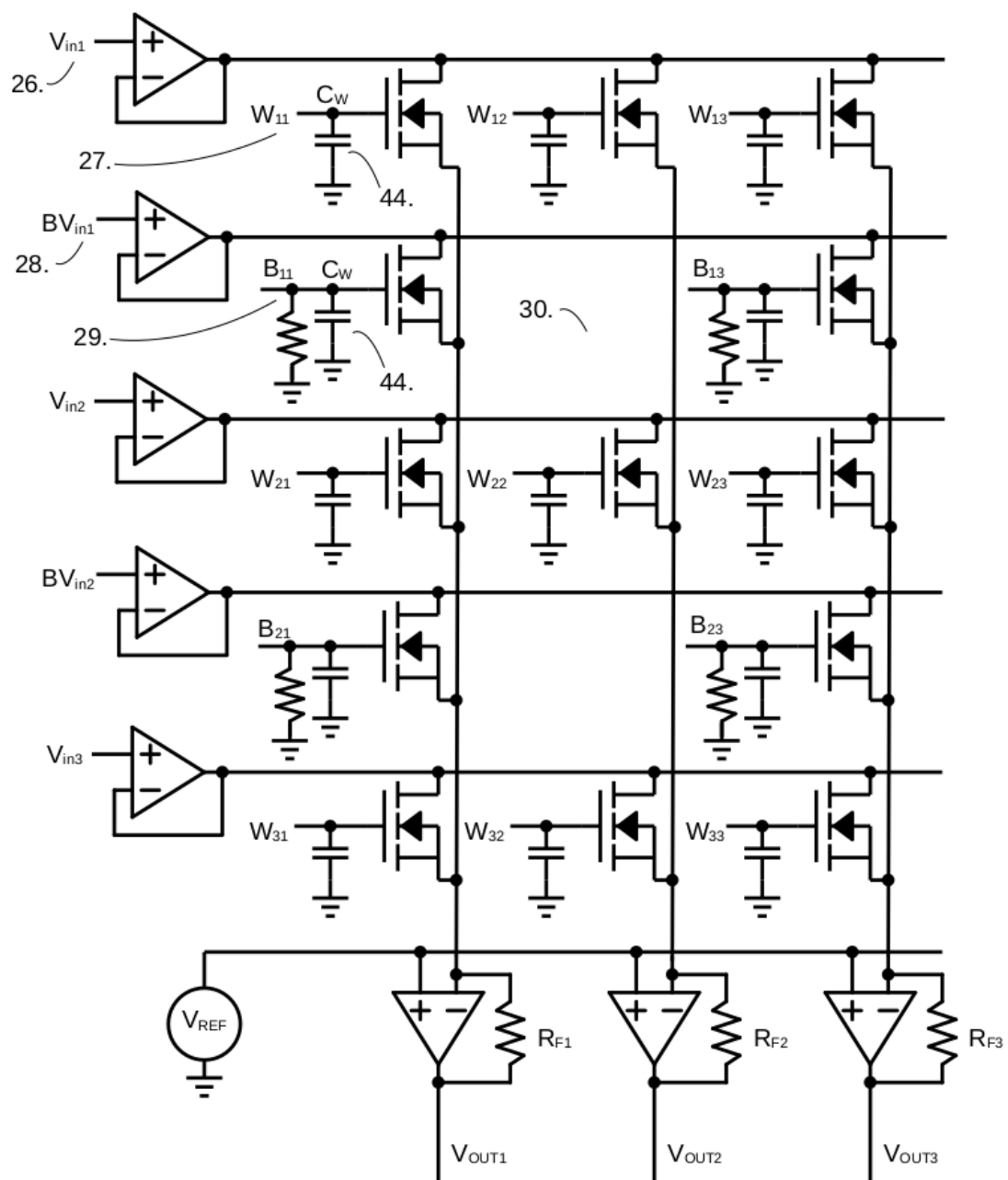


Fig 2

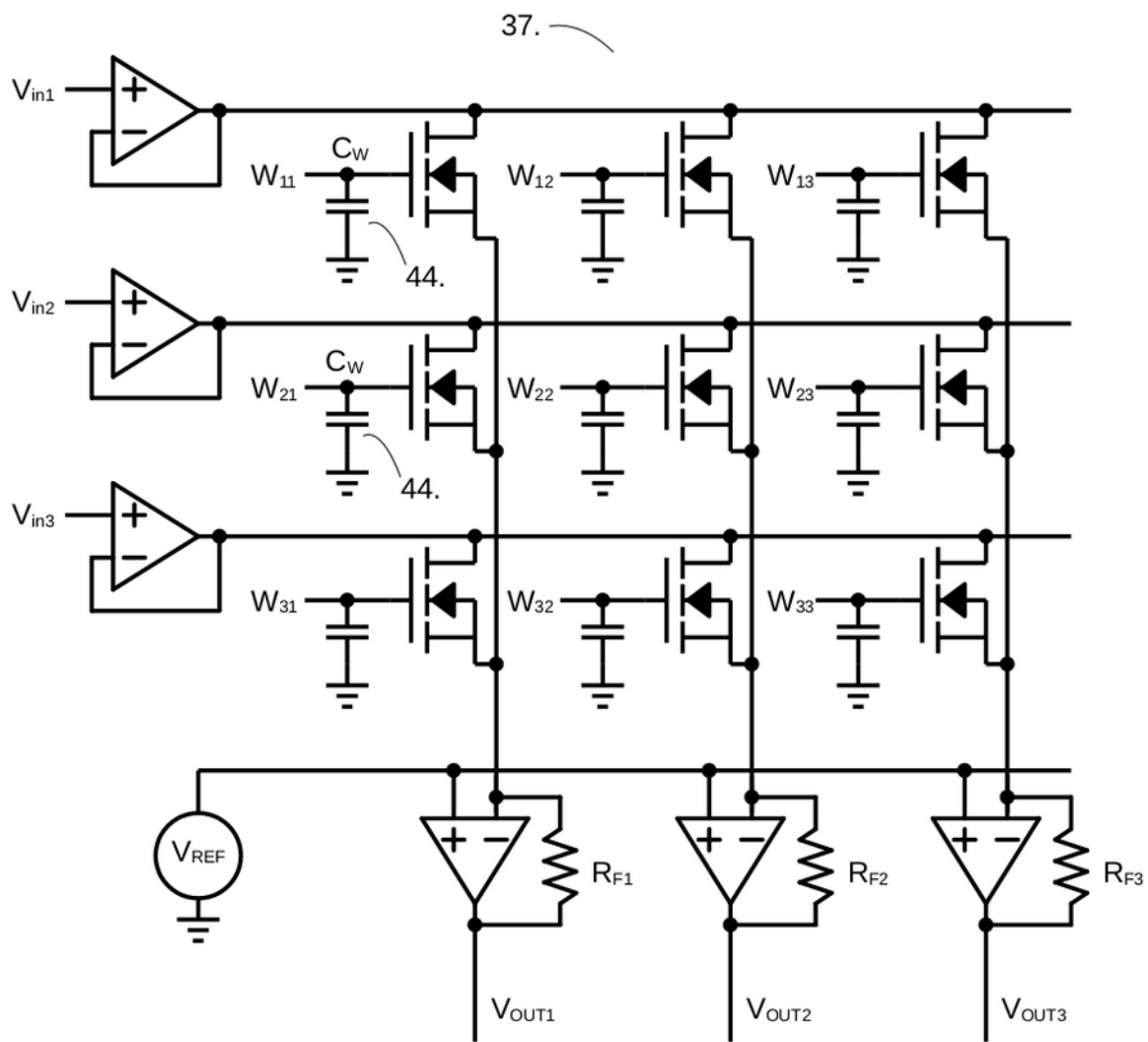


Fig 3

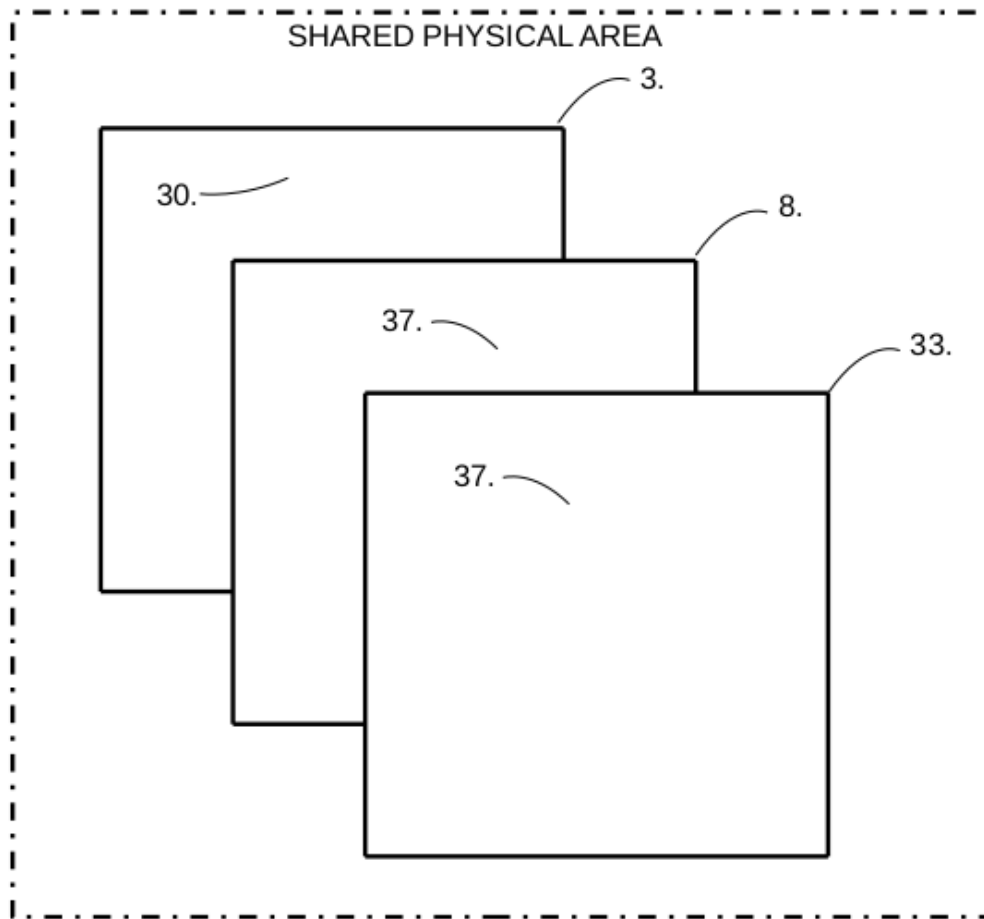


Fig 4

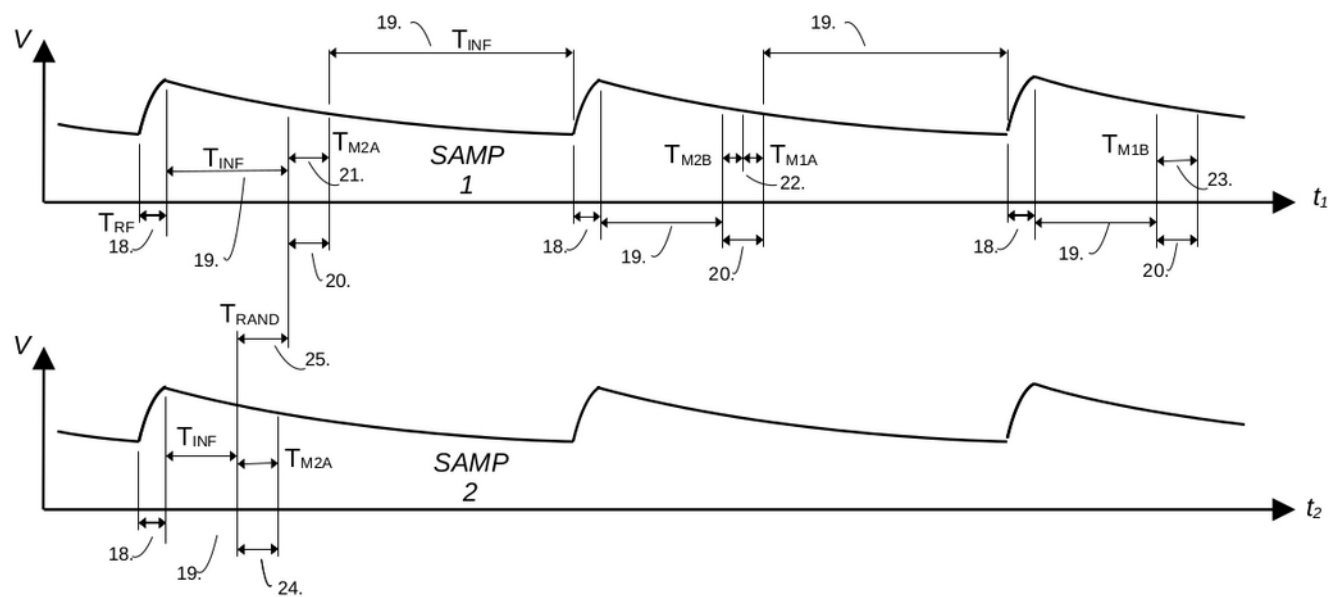


Fig 5

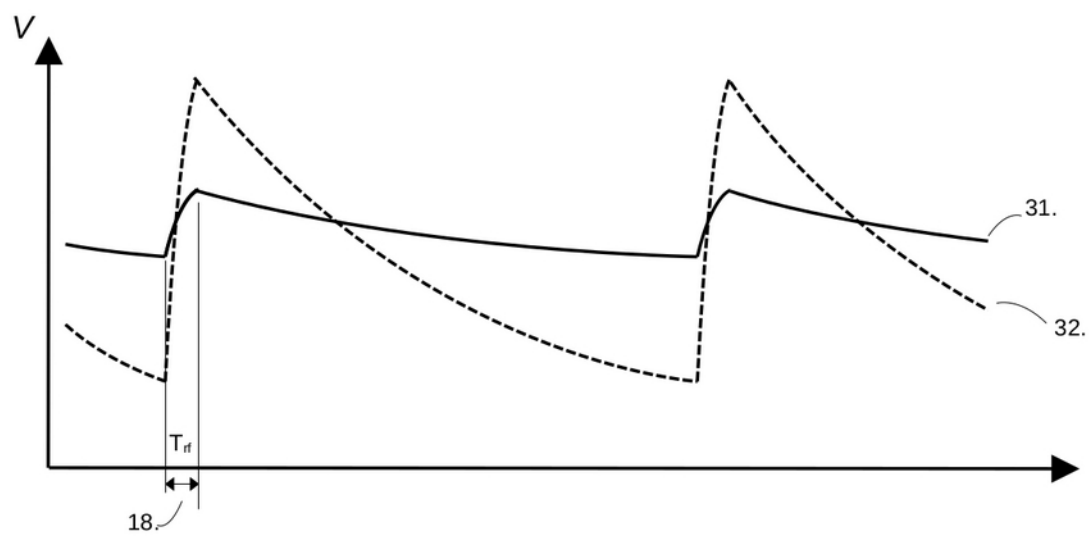


Fig 6

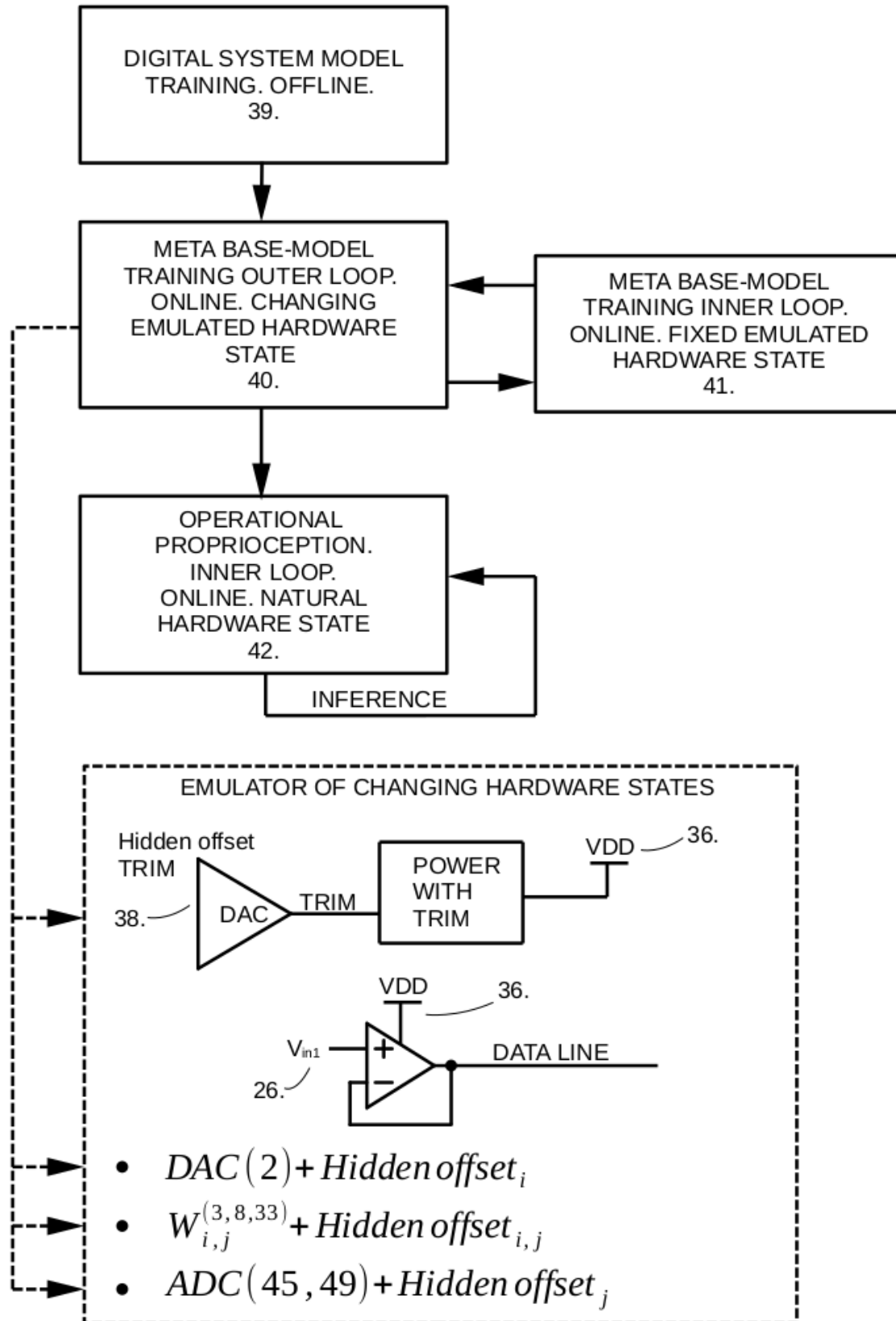


Fig 7

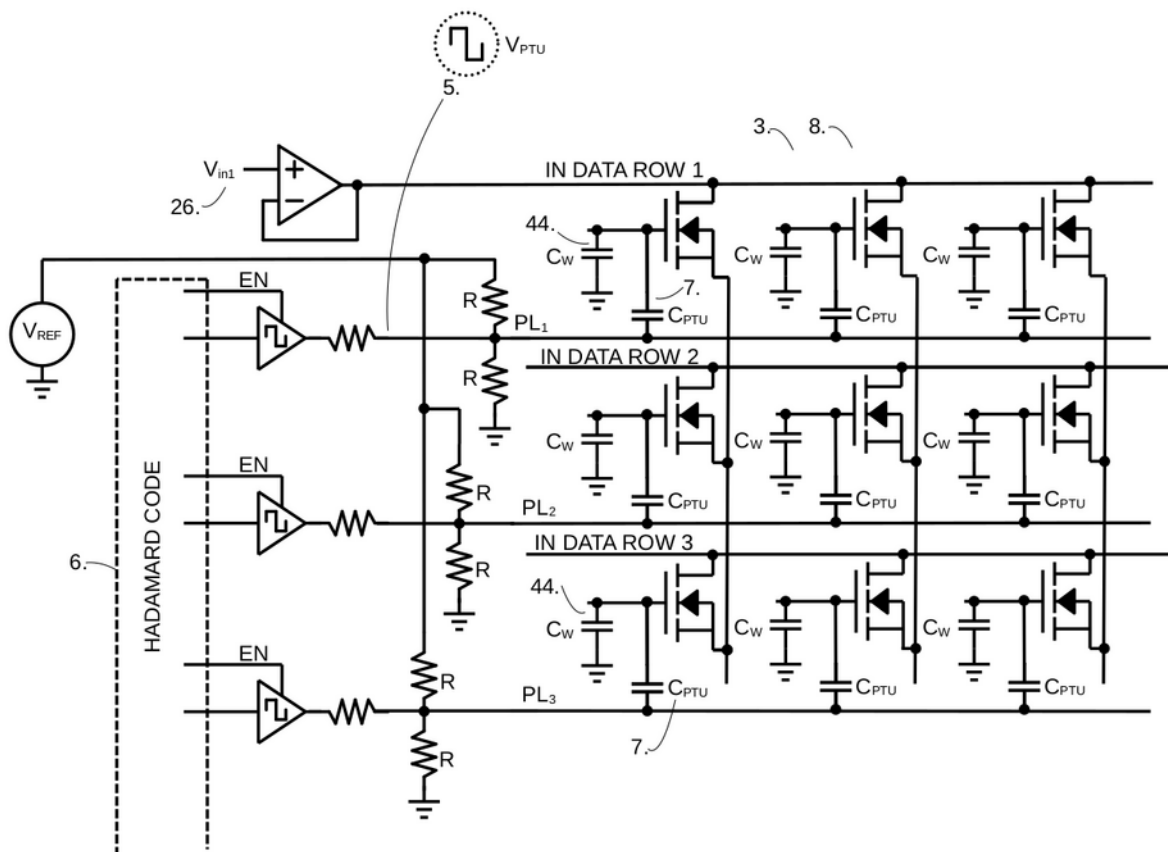


Fig 8

Line	T1	T2	T3	T4	T5	T6	T7	T8
1	+1	+1	+1	+1	+1	+1	+1	+1
2	+1	-1	+1	-1	+1	-1	+1	-1
3	+1	+1	-1	-1	+1	+1	-1	-1
4	+1	-1	-1	+1	+1	-1	-1	+1
5	+1	+1	+1	+1	-1	-1	-1	-1
6	+1	-1	+1	-1	-1	+1	-1	+1
7	+1	+1	-1	-1	-1	-1	+1	+1
8	+1	-1	-1	+1	-1	+1	+1	-1
Line	T9	T10	T11	T12	T13	T14	T15	T16
1	-1	-1	-1	-1	-1	-1	-1	-1
2	-1	+1	-1	+1	-1	+1	-1	+1
3	-1	-1	+1	+1	-1	-1	+1	+1
4	-1	+1	+1	-1	-1	+1	+1	-1
5	-1	-1	-1	-1	+1	+1	+1	+1
6	-1	+1	-1	+1	+1	-1	+1	-1
7	-1	-1	+1	+1	+1	+1	-1	-1
8	-1	+1	+1	-1	+1	-1	-1	+1

Fig 9 (43).

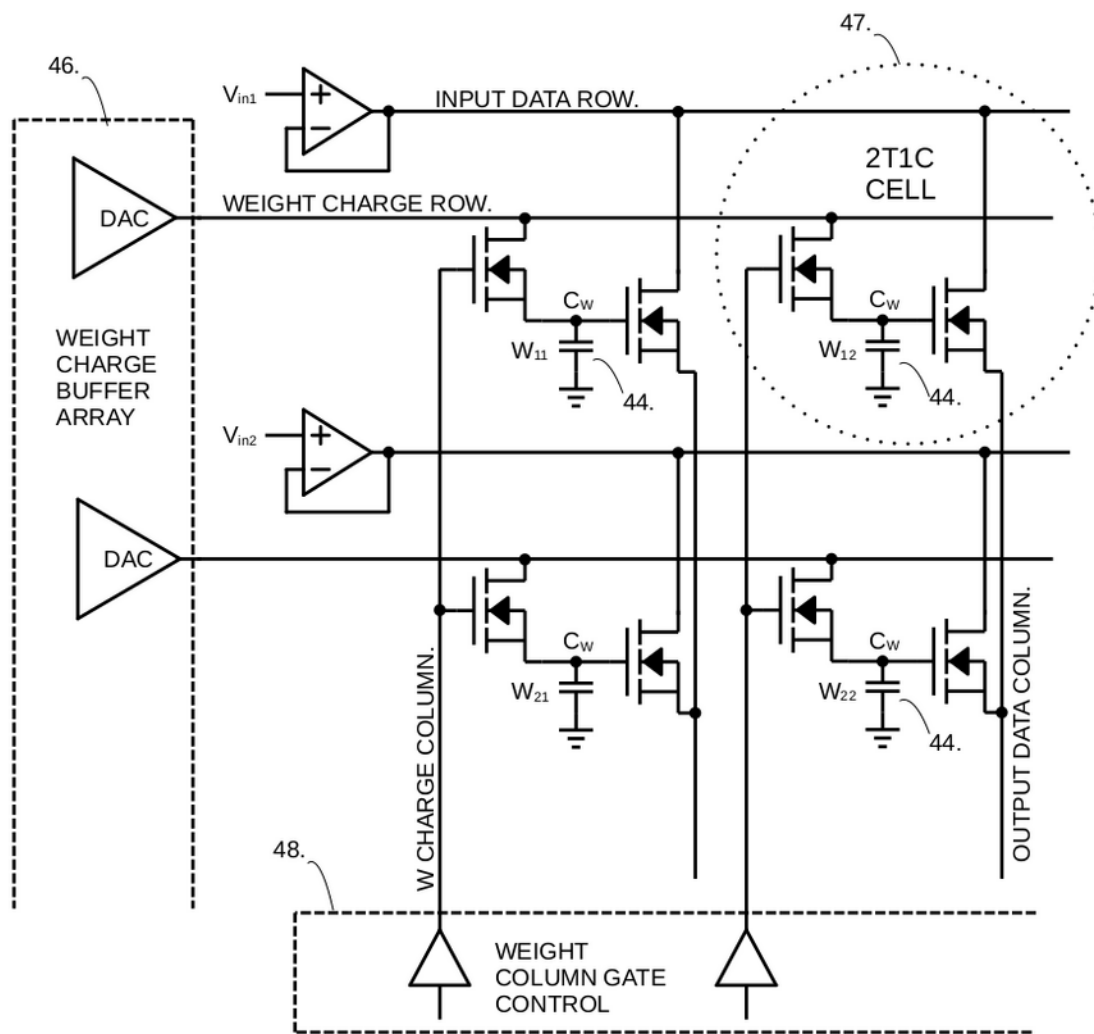


Fig 10

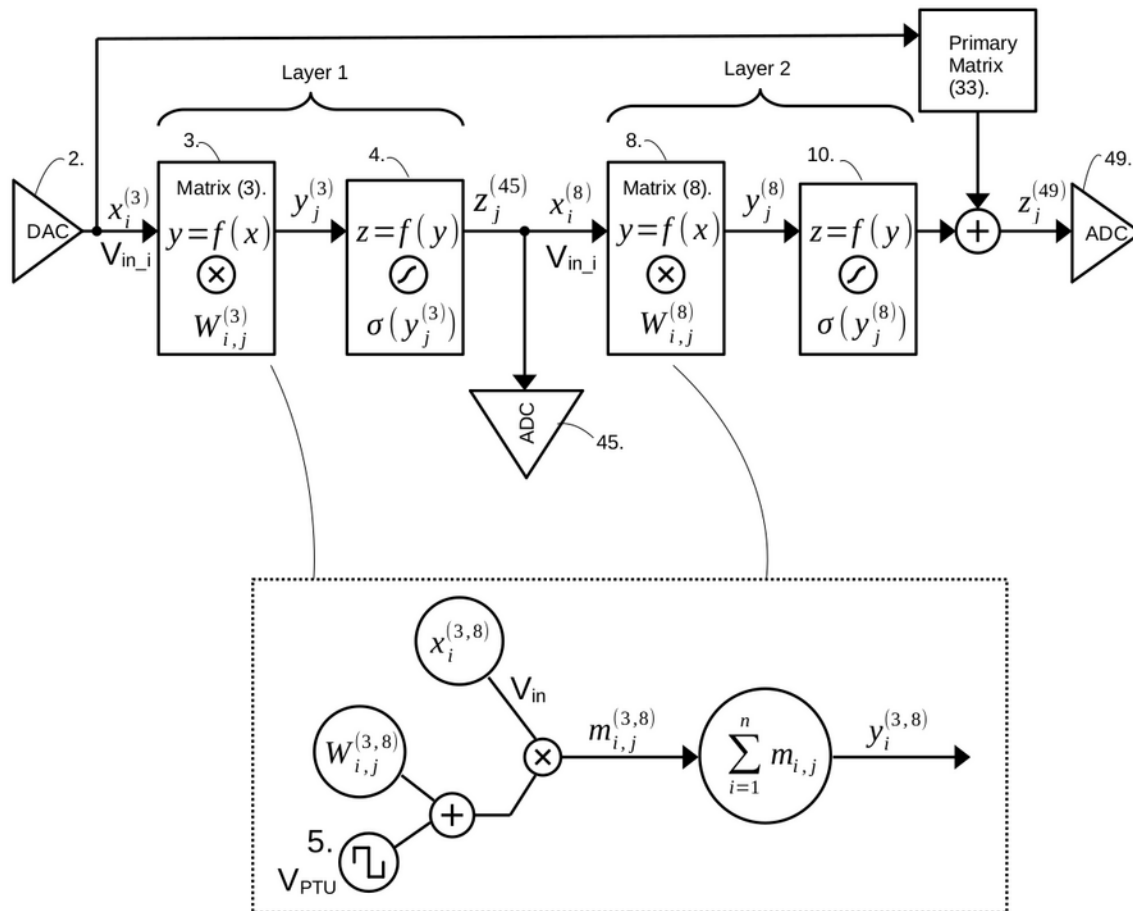


Fig 11

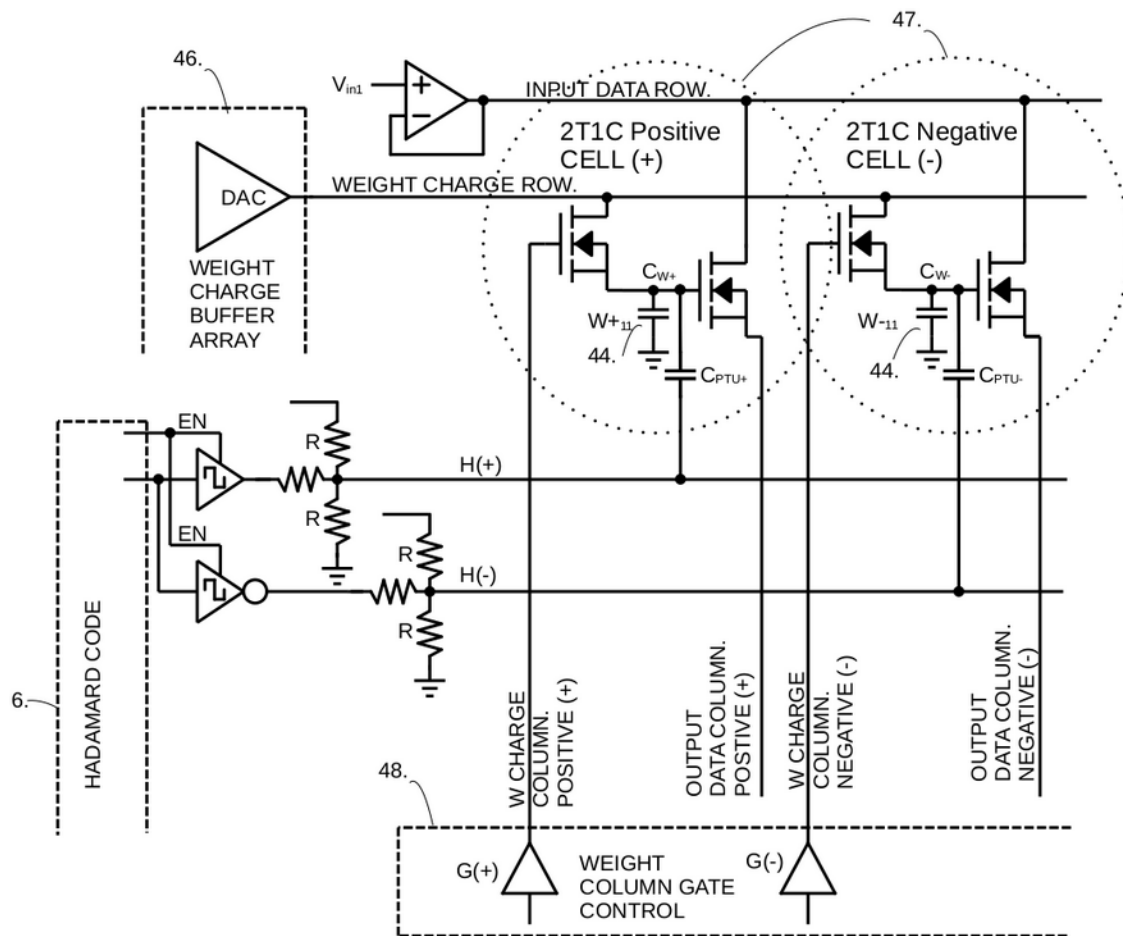


Fig 12

Detailed Description of the Invention

(Fig. 1) illustrates the asymmetry in the system, where the dashed lines (12) to (17) and (34) represent an intermittent control signal path for meta-learning-based calibration, which operates at a significantly lower bandwidth than the analog computation path marked with solid lines.

The digital control unit (9) communicates correction values via the control paths (12, 15, 34). It should be understood that these paths include, or are connected to, a digital-to-analog conversion unit (DAC) and a buffer array (46) for the purpose of updating the charge in the analog memory cells (47) and their storage capacitors (44).

(Fig. 10) illustrates a possible weight charging matrix with 2T1C cells (2 Transistor, 1 Capacitor). There, a buffer DAC array (46) charges the weight rows of the matrix with an analog weight voltage, while a digital column control driver (48) controls gate 'G' (the gate of the transistor that charges the weight) of an entire column of charging transistors for selective programming. The control paths (16, 36) are coupled to rows in the matrix (3) via a digital-to-analog converter (2) for input data.

System Architecture and Signal Flow.

To realize the inverted MAML architecture, the hardware is arranged in a specific sequential chain that enables both high-performance inference (19) and precise gradient extraction. A digital input signal (1) is converted in the DAC (2) and fed into the first analog correction matrix (3) (Layer 1), the output signal of which passes through a first activation function (4) (analog or digital). This signal is then fed into the second analog correction matrix (8) (Layer 2), followed by an activation function (10), creating an internal deviation correction signal up to the summer (35). In parallel, the main computation is performed from the input signal (2) into the analog matrix (33). The output signal from the primary matrix (33) is summed with the output signal from the activation function (10), which is then converted at full speed in an ADC (49) to create the digital output signal (11).

This atomic triad of matrices (Fig. 1) is arranged for continuous state monitoring via a digital control unit (9). To enable differential perturbation analysis with a high signal-to-noise ratio (SNR) without affecting the integrity of the stored analog weights, the system includes a modulated perturbation source (5). (V_{PTU}). The perturbation source is controlled by the control unit (9) and is connected to an encoding unit (6) (Hadamard encoder). Unlike a conventional selector, the encoding unit (6) is arranged to simultaneously inject controlled perturbation signals onto all row conductors (PL) in matrix (3) or (8). This injection takes place in accordance with an orthogonal coding scheme (Fig. 9) (43), where each individual row is assigned a unique pattern of phase positions (positive or negative polarity) over a measurement cycle. The injection occurs via dedicated perturbation capacitances C_{PTU} (7) whose capacitance values are chosen to be orders of magnitude lower than the actual storage capacitance of the weights C_W (44). This relationship, where C_{PTU} (7) < C_W (44), ensures that the encoded perturbation signals can modulate the row signals sufficiently for digital detection at the output, while the remaining charge (the weight) in the matrix remains unaffected. By utilizing orthogonal encoding, the control unit (9) can mathematically separate the summed output signal into individual partial derivatives

$$(\partial)$$

for each cell, even though all rows are excited simultaneously during the measurement window.

To maintain the charge of the weights over time and enable the iterative, inverted meta-learning adjustment, the system includes an addressing device (12, 15, 34). This configuration allows the control unit (9) to update weight values (27, 29) in the matrices based on the perturbation analysis extracted during the short measurement window (20), without interrupting the majority of the time during the discharge phase or disturbing the main high-frequency signal flow (payload) during inference (19). It should be noted that the primary computation matrix (33) does not include said perturbation signal mechanism, since the weights in the matrix (33) are exclusively redistributed (reallocated) but never adjusted in substance. Thereby, the need for perturbation analysis for this specific primary matrix (33) is eliminated.

The activation circuits (4, 10) and the summer (35) preferably consist of analog circuits to minimize energy consumption and complexity, but may alternatively be digital.

By utilizing analog operational amplifiers for the activation functions (4) and (10), a natural, smooth S-curve is obtained completely without the need for complex digital computational logic. Since the inverted MAML architecture provides continuous self-calibration, the inherent problems of component drift and unstable curve shapes are eliminated, which allows the analog saturation curve to be used as a stable and precise mathematical operator.

The control unit (9) is arranged to continuously compare the system's digital output signal (11, 13) with a digital reference model, which represents the deterministic truth. Hereby, the system is calibrated against the actual physical reality rather than against a theoretical simulation model, ensuring determinism even under changing environmental conditions such as fluctuating temperature.

In a preferred embodiment of the invention, the matrix unit (3) comprises at least one temporal reference cell (29). In contrast to the computing cells (27) of the matrix, which are optimized to maintain an electrical charge (31) over time, the temporal reference cell (29) is configured with a dedicated and accelerated discharge path (28). In one embodiment, this discharge path (28) may consist of a resistive element arranged in parallel with the storage capacitance of the cell, or in an alternative embodiment, of an actively biased transistor functioning as a constant-current source for controlled leakage.

The purpose of the discharge path (28) is to generate a discharge curve (32) with a steepness that substantially exceeds the natural voltage leakage (31) of the computing cells (27). By dimensioning the discharge path (28) such that the voltage drop (32) of the temporal reference cell covers a substantial part of, and preferably the entirety of, the available analog signal range (Full Swing) between two consecutive measurement windows (18), a physical normalization of the time axis is achieved.

This configuration maximizes the signal-to-noise ratio (SNR) during the digital control unit's reading, as the time information is correlated (mapped) against the full dynamic range of the matrix. The digital control unit can thereby, with high precision, use the curve (32) as an analog timestamp to interpolate and compensate for the significantly slower drift in the computing cells of the matrix via the inverted meta-learning algorithm.

The analog computing cores (30, 37) (Fig. 2), (Fig. 3) are constructed as a crossbar matrix where the input signals are injected via a set of input buffers (26). These buffers, preferably implemented as high-impedance operational amplifiers, ensure that the voltage signal V_{in} is driven stably across the rows of the matrix regardless of the combined resistive load from the programmable weights. By utilizing these precision buffers (26), initial signal errors are minimized already at the input stage, which provides the MAML algorithm with a more linear base to work from.

The system's matrix architecture is defined by two internal configurations: a reference structure (30) (Fig. 2) comprising discharge cells (29), and a computing structure (37) (Fig. 3) without discharge cells. The first matrix (3) is implemented as a hybrid structure where an upper part comprises the reference structure (30) and a lower part comprises the computing structure (37). The second matrix (8) and the third matrix (33) are, in their entirety, uniformly constructed according to the computing structure (37). Because the matrices (3, 8, 33) are arranged in thermal cooperation, the discharge cells (29) in the first matrix (3) are utilized as a common reference for the calibration of all matrices, allowing matrix (8) and (33) to be optimized for maximum computational area without internal reference cells.

A trimable DAC (38) for the supply voltage (36) V_{DD} for the triad of matrices (3, 8, 33) is available to emulate voltage variations during base model training.

Descriptions.

I. Gradient Extraction during the Measurement Window

II. Calibration Cycle, Meta-Adaptation.

I. The system's procedure for gradient extraction is performed in a sequential order as follows:

A digital input signal (1) is fed via the DAC (2) through the first matrix (3) and the second matrix (8) for "inverted" MAML. The input signal (1) via the DAC (2) is also fed through the primary computation matrix (33), which has been loaded with fixed, deterministic weights (34) represented from the digital ideal original model (39). The output signals from the activation function (10) are summed with the output signals from the primary matrix (33). The control unit (9) reads (samples) a reference value at the intermediate ADC (45) as well as the resulting error at the final output (11, 13).

The perturbation analysis takes place in two separate steps for the weights in matrix (3) and matrix (8) respectively. The measurement is done internally with the ADC (45) between Layer 1 and Layer 2. This is necessary to correctly calculate the chain rule digitally, and to avoid "spaghetti" mixed columns and rows between matrix (3) and matrix (8) in the perturbation analysis.

Analysis of the Chain Rule and Total Loss for Matrix (8):

To extract the local derivative for an individual cell (weight) in matrix (8) in relation to the total loss function (Total Loss), the control unit (9) applies a hardware-assisted calculation of the chain rule.

When the encoding unit (6) is active, a known perturbation $V_{PTU}(5)$ is created on (PL-LINE). A momentary dynamic change (AC delta) then occurs, which propagates to the output. The control unit measures this AC delta at each column after the activation function (10).

The calibration process is executed in three distinct steps to maximize precision and system utilization:

1. Data Collection (Excitation Phase): During a defined temporal measurement window (20), the Hadamard-encoded (Fig. 10) perturbation signals are injected simultaneously onto the rows via the encoding unit (6). The resulting summed analog output signal is sampled by the ADC (49) or at the internal ADC (45). Once the measurement series for the orthogonal sequence is complete, the measurement window (20) closes.

2. Digital Extraction (FHT Phase): Immediately after the measurement window closes, the digital control unit (9) processes the stored measurement values with a Fast Hadamard Transform (FHT). This mathematical decoding separates the summed signal into individual partial derivatives

(∂)

for each specific cell.

3. Adaptation (Chain Rule Calculation): The isolated derivatives

(∂)

are subsequently used in the chain rule calculations to determine the necessary weight adjustments. It should be noted that while step 1 requires physical access to the matrix's signal path, steps 2 and 3 take place entirely in the digital domain in parallel with the system's ordinary inference (19) continuing. This asymmetric flow enables continuous calibration without compromising the system's real-time performance or latency.

The partial derivative for each individual cell intersection (weight) in matrix (8) is calculated in three cooperating steps that utilize the parallelly extracted data:

1. Global Error Determination: The digital system evaluates the derivative of the global loss

$$\frac{\partial Loss}{\partial Out}$$

at the final output (11). This constitutes the starting point for the chain rule.

2. Local Error Propagation and Analog Weighting (Fig. 11):

Instead of using a raw measurement value, the partial derivatives isolated via the Fast Hadamard Transform (FHT) after the measurement window (20) are utilized to obtain a unique error gradient for each specific cell intersection. As illustrated in (Fig. 11), the subsequent weighting against input data $x_i^{(8)}$ occurs completely natively and automatically through the matrix's analog multiplication. By combining the Hadamard-decoded derivatives with this automatic analog multiplication and the global error, the system can execute a complete backward propagation calculation ("Backpropagation") directly in the hardware.

Equations for Gradient Extraction during the Measurement Window.

i = matrix row

j = matrix column

Gradient Extraction for Matrix (8):

Equation 1

$$L_{total} = \sum_{j=1}^n \left(Er_j^{(49)} \right)^2$$

also represented as:

$$L_{total} = \left(Z_{target} - Z_{out} \right)^2$$

Equation 2

Extraction of weight derivatives (delta) for matrix (8):

$$\frac{\partial L_{total}}{\partial W_{i,j}^{(8)}} = \left\langle \frac{\partial z_j^{(49)}}{\partial V_i^{PTU(5)}} \right\rangle_{\text{Hadamard}} \cdot Er_j^{(49)}$$

It is noted that Equation 2 excludes the explicit multiplication with the row's input signal $x_i^{(8)}$, which is characteristic of conventional digital backpropagation. This unique mathematical simplification is a direct technical effect of the system's analog transistor dynamics at the crossbar intersection (i, j) . Because the transient perturbation — hereinafter referred to as the weight perturbation (weight shaking) $\partial V_i^{PTU(5)}$ — is physically applied via the row line, the propagation of the shaking to the column line is momentarily modulated by the cell's prevailing conductance and the row's analog voltage level. If, for example, the input signal on a row is zero $x_i^{(8)} = 0$, the weight perturbation (weight shaking) is completely attenuated in the physical cell intersection and generates no contribution to the column's analog summation line.

The measurement value extracted via the Hadamard decoding

$$\left\langle \frac{\partial z_j^{(49)}}{\partial V_i^{PTU(5)}} \right\rangle_{\text{Hadamard}}$$

is thus not an isolated partial derivative for the weight

alone, but already carries an implicit, hardware-embedded multiplication with the row's input signal.

Because the multiplication occurs natively, automatically, and "for free" in the analog medium, the complete and correct gradient for matrix (8) is obtained by the control unit (9) digitally multiplying the Hadamard-deciphered measurement value directly by the global column error $Er_j^{(49)}$.

To digitally perform an additional multiplication with $x_i^{(8)}$ at this stage would constitute a mathematical double-counting (squaring of the input signal) and destroy the algorithm's convergence.

Where:

$$\partial W_{i,j}^{(8)}$$

represents the derivative of an individual cell's weight in matrix (8).

$$\partial V_i^{PTU(5)}$$

representerar den injicerade störningen från kodningsenheten (6) som genererar V_{PTU} (5) the input signal on a row i into matrix (8).

$$Er_j^{(49)}$$

represents the static global column error obtained from the output ADC (49) compared against an ideal digital target vector.

$$\partial z_j^{(49)}$$

represents the AC variation of an entire column measured from the ADC (49), corresponding to columns in matrix (8) that have propagated through the activation function (10) and summation unit (35).

Optimization Objectives and Error Correction

To enable the inverted meta-learning algorithm to minimize the deviation from the analog output Z_{OUT} and the digital reference Z_{TARGET} , a loss function is defined. The control unit (9) uses the Mean Squared Error (MSE) method to define this error. The squaring ensures that large deviations are prioritized in the calibration and that the error value is always positive, which is a prerequisite for the algorithm's convergence.

$$L_{total} = (Z_{target} - Z_{out})^2$$

Equation 3 Matrix (8) (theoretical). For backpropagating the derivative (delta) to matrix (3). Row-wise incoming derivative (delta) from matrix (8) (on the input side):

$$\frac{\partial x_i^{(8)analog}}{\partial x_i^{(8)}} = \sum_{j=1}^n W_{i,j}^{(8)analog} \cdot \sigma'(y_j^{(8)}) \cdot Er_j^{(49)}$$

represents the ideal, but in practice unknown, analog sensitivity of row i in the physical medium.

Where:

$$\frac{\partial L_{total}}{\partial x_i^{(8)}}$$

represents the derivative (delta) from matrix (8) (on the input side of matrix (8)) when the control unit (9) performs a perturbation on matrix (8).

Where:

$$y_j^{(8)}$$

represents the output signal of columns from the matrix multiplications in matrix (8) before these columns enter the activation function (10).

Equation 4 Matrix (8) (practical implementation). Row-wise incoming derivative (delta) from matrix (8) (on the input side):

$$\frac{\partial x_i^{(8)}}{\partial x_i^{(8)}} = \sum_{j=1}^n W_{i,j}^{(8)digital} \cdot \sigma'(y_j^{(8)}) \cdot Er_j^{(49)}$$

$\frac{\partial x_i^{(8)}}{\partial x_i^{(8)}}$ represents the digitally stored or calculated sensitivity parameter for row i into said second matrix (8), which constitutes a digital value "receipt" of how a differential transient change on the input row propagates to the output.

$$W_{i,j}^{(8)digital}$$

represents the digital target value for the weight (conductance value) in the cell junction of row i and column j of said second matrix (8).

$$\sigma'(y_j^{(8)})$$

represents the extracted instantaneous, non-linear analog slope (derivative) of the activation function (10) for column j .

To determine said analog slope, $\sigma'(y_j^{(8)})$ without physical perturbation injection on all row lines, and to eliminate noise amplification due to input signals close to zero, the digital control unit (9) is arranged to execute a selective carrier wave search. The control unit (9) evaluates the stored digital values for the input signal vector $X^{(8)}$, which have been sampled by the intermediate ADC unit (45) during a forward calculation pass (Forward Pass).

The control unit (9) identifies a specific row index k (also referred to as the maximum carrier wave row $X_{imax}^{(8)}$) which exhibits the highest instantaneous signal amplitude. The hidden analog column derivative is then extracted entirely digitally by multiplying the differential AC response, isolated via the Hadamard decoding for the selected row k , with the inverse of its known static carrier wave amplitude according to:

Equation 5: (theoretical)

$$\sigma'(y_j^{(8)}) = \frac{1}{X_k^{(8)}} \cdot \left\langle \frac{\partial Z_j^{(49)}}{\partial W_{k,j}^{(8)}} \right\rangle_{\text{Hadamard}}$$

Because said activation function (10) and its associated analog summation line are common to all memory cells in column j , the digitally extracted analog slope is identical for the entire column. The control unit (9) is arranged to mirror the calculated value for $\sigma'(y_j^{(8)})$ as a constant multiplier for all rows i during the execution of Equation 3, thereby eliminating the need for physical transient injectors on the intermediate row lines.

To clarify the physical mechanism in the matrix junction, where the deciphered measurement carries an implicit input signal component from the carrier wave row, the Hadamard-decoded measurement value is decomposed according to:

Explanatory auxiliary equation (Equation 5a):

$$\left\langle \frac{\partial z_j^{(49)}}{\partial W_{k,j}^{(8)}} \right\rangle_{\text{Hadamard}} = \sigma'(y_j^{(8)}) \cdot x_k^{(8)}$$

By substituting Equation 5a into Equation 5, the exact mathematical elimination (canceling out) of the row signal is achieved:

Explanatory auxiliary equation (Equation 5b):

$$\sigma'(y_j^{(8)}) = \frac{1}{x_k^{(8)}} \cdot [\sigma'(y_j^{(8)}) \cdot x_k^{(8)}] = \sigma'(y_j^{(8)}) \cdot \frac{x_k^{(8)}}{x_k^{(8)}} = \sigma'(y_j^{(8)})$$

Description of the Virtual Row Perturbation via Implicit Hardware Elimination:

The physical significance of Equation 5 and Auxiliary Equation 5a is fundamental to bridging the asymmetry between the network's forward signaling and its adaptive backpropagation. Since the system utilizes a physical forward perturbation (Forward Perturbation) via AC injection on the selected carrier wave row k , the transistor dynamics of the cross-point in cell (k, j) will automatically multiply the applied voltage variation by the prevailing stationary row signal $x_k^{(8)}$. The measured value read out via the differential measurement window and the Hadamard transform is thus not a pure partial derivative with respect to the weight alone, but instead contains the hidden composite product $\sigma'(y_j^{(8)}) \cdot x_k^{(8)}$ as shown in Equation 5a. To isolate the pure, stripped down "naked" analog slope σ' of the activation function (10) — which is unique to the column but universal for all its connected rows — the control unit (9) executes the digital correction step in Equation 5. By multiplying the composite measured value by the inverse of the carrier wave's known digital value $\frac{1}{x_k^{(8)}}$, the

hardware-based multiplication is mathematically canceled out. Since said activation function (10) and its associated analog summation line are common to all memory cells in column j , the analog slope digitally

isolated in this manner is identical for the entire column.

The control unit (9) is arranged to reflect the pure value thus calculated for σ' as a constant multiplier for all rows during the execution of Equation 4.

By multiplying this isolated column slope with the stored digital matrix weights in a subsequent digital stage, $W_{i,j}^{(8)digital}$, is the effect of a row perturbation (row shaking) virtually reconstructed at $\partial x_k^{(8)}$. This unique interaction between physical weight perturbation (weight shaking) and digital inverse scaling completely eliminates the need to integrate physical, transient injector circuits or dedicated AC coupling capacitors on the intermediate analog row lines between matrix (3) and matrix (8), which radically reduces the silicon area and lowers the system's total parasitic capacitance.

Equation 6: (practical implementation, replaces equation 5 in practice)

$$\sigma'(y_j^{(8)}) \approx \frac{x_k^{(8)}}{(x_k^{(8)})^2 + \alpha} \cdot \left\langle \frac{\partial z_j^{(49)}}{\partial W_{k,j}^{(8)}} \right\rangle_{\text{Hadamard}}$$

The parametric regularization factor α constitutes a static, strictly positive digital constant stored in the memory of the control unit (9). The factor α is dimensioned to function as a numerical floor protection (threshold value) to prevent digital overflow and system crashes resulting from division by zero. The value of α is chosen to be small enough not to affect the computational accuracy during normal operation (when the carrier amplitude is strong), but large enough to dominate the denominator and attenuate the expression during zero-crossings or extreme noise situations. In a preferred embodiment, α is configured within the order of magnitude of 10^{-6} to 10^{-4} , preferably 10^{-5} . At a zero-crossing where $x_k^{(8)} \rightarrow 0$, forces α the denominator to stabilize at a finite, deterministic value instead of collapsing toward zero. The entire fractional expression thus safely assumes the value of zero, which effectively blocks unrestricted noise amplification and guarantees that the iterative meta-learning algorithm remains stable during real-time execution.

Later, the saved $\partial x_i^{(8)}$ is used when the control unit (9) turns off the Hadamard-coded perturbation on matrix (8) and the control unit (9) instead isolates the Hadamard-coded perturbation on matrix (3). The control unit (9) then uses this $\partial x_i^{(8)}$ as a sensitivity value for this row on matrix (8)

instead of reading the output total loss $\frac{\partial L_{total}}{\partial x_i^{(3)}}$ when the control unit (9)

has the perturbation analysis enabled against matrix (3).

It should be noted that the calculation of $\partial x_i^{(8)}$ is based on the known digital target matrix $W_{i,j}^{(8)digital}$, which implies that any deviations in the actual gain (gain error) of the individual analog cell in relation to the digital value in practice only manifest as a local, cell-specific variation in learning rate. Thanks to the system's closed feedback loop against the global static error $Er_j^{(49)}$ the architecture is inherently robust against such individual gain deviations, and the system's total convergence is achieved without the exact, prevailing gain of the subsequent analog layer needing to be measured or known.

This method of splitting the perturbation analysis into two separate parts, matrix (8) followed by matrix (3), reduces rounding errors and noise (SNR) compared to running the entire chain simultaneously in a single, joint perturbation analysis measurement.

Isolation and analysis of the first layer (Matrix 3)

The analysis of the first layer (matrix 3) is performed in parallel and asymmetrically to maximize the signal-to-noise ratio (SNR) and avoid error propagation. Instead of sequential scanning, the Hadamard encoder (6), which generates the V_{PTU} (5) signal, injects a known differential perturbation simultaneously on all rows in matrix (3) using orthogonal patterns. The resulting variations in the matrix's output signal are measured immediately at the intermediate ADC (45) as local column-specific AC deltas $\partial z_j^{(45)}$.

By applying a Fast Hadamard Transform (43) (FHT) in the control unit (9), the individual derivatives for each specific row/column intersection can be isolated from the aggregated signal. This measured value captures asymmetries and nonlinearities in both the analog crosspoint and the associated activation function (4). To extract the specific derivative for an individual cell in matrix (3) with respect to the final loss function (Total Loss), the control unit (9) applies a digital-to-analog chain rule based on saved sensitivity data:

1. Retrieval of saved sensitivity: The control unit (9) retrieves the digitally stored sensitivity parameter $\partial x_j^{(8)}$ for matrix (8). This value constitutes a static "receipt" of how sensitive the subsequent layer is to changes on its inputs.

2. Local gradient extraction: The physically measured AC delta for each column at the ADC (45) is deciphered via FHT to obtain the local derivative

$$\left\langle \frac{\partial z_j^{(45)}}{\partial V_i^{PTU(5)}} \right\rangle_{\text{Hadamard}}.$$

3. Analog input modulation and gradient determination (Fig 11):

The modulation with the input signal from the DAC (2) occurs inherently and automatically through the analog multiplication of the matrix (3), as illustrated in (Fig 11). Through this integrated interaction between the input signal, the deciphered internal derivative, and the system's sensitivity, the exact derivative for each specific crosspoint in matrix (3) is obtained. The method ensures that the training of the first layer remains strictly deterministic and immune to instantaneous drift or noise in subsequent layers.

For matrix (3), the output columns switch to rows in matrix (8), thus:

$$z_j^{(45)} = x_i^{(8)}$$

thus the indexing also switches over from i to j even for

$$\partial x_i^{(8)} \rightarrow \partial x_j^{(8)}$$

Note that this $\partial x_i^{(8)}$ is retrieved from a saved measurement when the perturbation measurement was performed in isolation on matrix (8), i.e., not when the perturbation measurement is occurring on matrix (3) in the equation below.

Equation 7

Extraction of weight derivatives (delta) for matrix (3):

$$\frac{\partial L_{total}}{\partial W_{i,j}^{(3)}} = \left\langle \frac{\partial z_j^{(45)}}{\partial V_i^{PTU(5)}} \right\rangle_{\text{Hadamard}} \cdot \partial x_j^{(8)}$$

Where:

$$\partial z_j^{(45)}$$

represents the entire column's AC variation measured from the internal ADC (45) between layer 1 and layer 2, which corresponds to columns in matrix (3) that have propagated through the activation function (4).

Where:

$$\partial V_i^{PTU(5)}$$

represents the injected perturbation from the encoder unit (6) generating the $V_{PTU(5)}$ signal into a row i of matrix (3).

Where:

$$\partial W_{i,j}^{(3)}$$

represents the derivative of an individual cell's weight in matrix (3).

II. Calibration cycle, Meta-adaptation:

With the stored derivatives from both correction matrices (3, 8), the control unit (9) performs a meta-learning-based update (40, 41) of the system's control parameters to minimize the global error at the output (11).

Detailed description: hybrid architecture with triple matrices and non-linear deviation correction The present invention utilizes an architecture that separates the logical computation from the physical hardware compensation through three interacting matrix units: The primary computation matrix (33):

This analog matrix is loaded with frozen, deterministic weights (34) from a digital source model (39) (e.g., an LLM). This unit performs the main mathematical operation but is subject to chip-specific physical non-idealities.

The dual MAML correction branch, the correction matrices (3, 8):

To handle complex, non-linear errors (such as XOR-like dependencies and S-shaped saturation curves), the correction branch comprises at least two serial matrix layers with intermediate non-linear activation functions. First stage (3): Projects the input signal (2) to a hidden space via a first correction matrix (3) and activation function (4). Second stage (8): Transforms the hidden representation via the correction matrix (8) and activation function (10) to generate a higher-order correction vector.

The output from the correction branch (from matrix 8) is routed to an analog activation function (10). Thereafter, the output signal from the analog activation function (10) is linearly summed with the output from the primary computation matrix (33) via an adder (35). Said adder (35) is preferably an analog summing circuit configured to maintain the signal processing entirely in the analog domain up to the summation point. This combined analog signal vector then undergoes a final analog-to-digital conversion (ADC), which ensures that the entire computation block delivers a digital result that corresponds to an ideal digital multiplication matrix.

Base model training via isolated hardware emulation (Fig 7).

According to one embodiment, the system is trained into a robust base model through a method based on controlled hardware emulation, wherein the physical hardware substrates (the atomic cores) are kept completely isolated from each other during the entire process. In contrast to methods based on spatial permutation or relocation of logical weights across separate substrates, each individual inverted MAML atom is arranged to exclusively operate with its own dedicated hybrid atomic triad consisting of the analog matrices (3, 8, 33).

To expose the base model to a sufficient search space during training, a dynamic emulation of thermal drift, voltage deviations, and altered discharge characteristics is introduced. In the outer loop (40) of the base model training, it is emulated that the three analog matrices (3, 8, 33) rapidly change their physical state. This emulates environmental effects such as heating or cooling of the hardware, accelerated or retarded discharge rates, or that the supply voltage V_{DD} (36) has drifted from its nominal value.

To emulate voltage drift, the hardware architecture comprises a dedicated trim-DAC (38), which is arranged to be able to shift the supply voltage V_{DD} (36) within a predetermined interval, preferably in the order of magnitude of $\pm 5\%$. The remaining part of the state emulation is realized programmatically via the digital control unit (9) by intentionally adding hidden offset values to the matrix weights $W^{(3,8,33)}_{i,j}$, as well as hidden, static digital offset values on the input signal's DAC (2) and the signal paths' ADC (45) and ADC (49). These offsets are applied without the inverted MAML algorithm having knowledge of or direct access to the actual offset values.

The digital control unit (9) is arranged to apply these hidden state shifts according to the following specific mechanisms:

1. Column-wise gain deviation: A joint offset is applied across an entire column of matrix weights, which for the inverted MAML algorithm emulates that the analog gain for the column in question has increased or decreased.
2. Time-based discharge deviation: An unknown offset is applied to the discharge time of the bias cells (29), which misleads the inverted MAML algorithm into interpreting the hardware state as though the discharge phase has progressed faster or slower than normal.

3. Input deviation (offset) and leakage emulation (DAC 2): A hidden static offset addition is applied to the digital input vector before it is converted by the DAC (2), which emulates a physical input offset voltage in the driver stages or constant electrical current leakage on the input lines.

4. Converter deviation and non-linear distortion (ADC 45 and ADC 49): By adding or subtracting fixed digital values after the analog-to-digital conversion in ADC (45) and ADC (49), DC offsets, thermal zero-point errors, or dark currents in the signal path's amplifier stages are emulated. This intentionally shifts the operating point of the signal in relation to the non-linear activation functions (4, 10), forcing the inverted MAML algorithm to develop a proprioceptive ability to handle and calibrate away complex non-linear hardware skewness.

These artificial variations are generated by the digital control unit (9) secretly adding the hidden offset pattern across selected weights and converter registers. This unique pattern is then kept strictly fixed during the execution of the MAML algorithm's inner loop (41) in order to emulate a stable but shifted hardware state.

To ensure that the base model becomes robust and is not overfitted to static input signals during this fixed operating state, the system is arranged to permute and vary a multiplicity of different incoming digital vectors via the DAC (2) during the inner loop (41). Through this procedure, the underlying characteristic pattern for a specific hardware individual is maintained, while its virtual state location is displaced during the base model training to correspond to normal field variations, thereby establishing an optimal search space for the algorithm.

Upon transitioning to operational proprioception and normal operation in the inner loop (42), the digital control unit (9) is arranged and configured to completely remove all emulated and hidden offset additions from weights

$W^{(3,8,33)}_{i,j}$, DAC (2), ADC (45), and ADC (49). Simultaneously, the trim-DAC (38) is reset to its nominal center position. The system thereafter operates under its natural hardware conditions, whereby the real-time operating inner calibration loop fine-tunes the correction matrices (3, 8) against the actual physical substrate to counteract instantaneous physical drift. Fig 12 illustrates a differential implementation of the analog matrix comprising dual 2T1C cells for a differential multiplier. To establish a virtual zero point, a differential mirror structure according to industry standard is utilized, where the synaptic weight is represented by a positive weight component ($W+$) and a negative weight component ($W-$).

Fig 12 specifically illustrates how the system's row perturbation (for example, a Hadamard perturbation) is integrated into this environment. To optimize the perturbation response, the system is arranged to apply the perturbation signal in a differential and inverted manner across this structure. More specifically, the perturbation injection comprises two separate perturbation rows; a first positive perturbation row applied to the positive weight component ($W+$) and a corresponding inverted (negative) perturbation row applied to the negative weight component ($W-$). This configuration of differential and inverted perturbation rows ensures that the resulting transient signal can be isolated and evaluated with high precision across the differential structure, with minimized impact from common-mode noise. It should be noted that the use of the weight components ($W+$, $W-$) and the mirrored structure constitutes an industry standard for establishing said virtual zero point. Since individual multiplier transistors can only conduct current in one direction, the virtual zero point is intentionally raised from absolute zero current in the column. Through the implementation of a mirrored dual column, a differential output signal is thus generated that is essentially insensitive to offsets from the virtual midpoint.

Patent Claims

Claim 1.

A system for analog computation of neural networks, comprising a matrix structure which comprises an analog primary computation matrix (33) and a pair of analog correction matrices (3, 8), which together form an atomic triad, and which comprise a plurality of analog memory cells (27) where weights are stored as electrical charge on capacitances (27), and a digital control unit (9) coupled to said matrices, characterized in that the digital control unit is arranged to execute a meta-learning algorithm (MAML) configured for internal hardware adaptation of said atomic triad, wherein said algorithm is arranged to identify and compensate for internal stochastic and temporal variations of the matrices by iteratively adjusting the weights in the correction matrices (3, 8) to minimize the deviation from a digital reference model (39), wherein the primary computation matrix (33) constitutes an analog representation of the weights of said reference model.

Claim 2.

The system according to claim 1, characterized in that said matrix structure (3, 8, 33) comprises a plurality of reference nodes (29) distributed in a regular network (30) across the total area (30, 37) of the matrices, wherein said nodes (29) provide local measured values of spatial and temporal variations to the digital control unit (9), enabling a spatially resolved mapping of thermal and process-related gradients to guide the meta-learning algorithm's adjustment of the correction matrices (3, 8).

Claim 3.

The system according to any of the preceding claims, characterized in that the system comprises a perturbation source (5) arranged to generate an oscillating voltage amplitude and an encoding unit (6) arranged to modulate said amplitude to inject controlled perturbation signals simultaneously into a plurality of rows in said correction matrices (3, 8) via perturbation capacitances C_{PTU} (7), wherein said encoding unit (6) is configured to control the phase or amplitude of the perturbation signals according to an orthogonal pattern, preferably Hadamard encoding (43), and wherein said perturbation capacitance (7) has a value substantially lower than the storage capacitance C_W (44) (44) for the weights of the memory cells, for the purpose of enabling parallel gradient extraction without affecting the integrity of the charge stored in the memory cells.

Claim 4

The system according to any of the preceding claims, characterized in that an analog-to-digital converter, ADC (45), is arranged between the first correction matrix (3) and the second correction matrix (8), wherein the control unit (9) is arranged to read (sample) the orthogonally encoded perturbation response from each respective matrix in parallel, wherein the control unit (9) is configured to, through digital deciphering, preferably Fast Hadamard Transform (FHT), isolate individual row- and column-wise gradients to digitally compute weight adjustments according to the chain rule, and wherein the computation for the first correction matrix (3) is performed by combining a local deciphered response at the ADC (45) with a digitally stored sensitivity parameter $\partial x_i^{(3,8)}$ representing the second correction matrix (8).

Claim 5.

A method for training a system according to claim 1, characterized in that the digital control unit (9), during a training phase, establishes a robust base model (40, 41) for the meta-learning algorithm through controlled physical hardware emulation of voltage variations and thermal variations, wherein the system, via a dedicated controllable digital-to-analog converter (trim-DAC) (38), is capable of shifting the supply voltage (36) within a predetermined interval during the outer loop (40) of the meta-learning algorithm, wherein said voltage shift is kept fixed during the inner loop (41) of the algorithm, in order to expose each isolated atomic triad exclusively to the physical variations of its own substrate.

Claim 6.

The system according to claim 1, characterized in that it comprises an addressing device (12, 15, 34, 46, 48) coupled to a central digital-to-analog converter (DAC) (46), wherein the control unit (9) is arranged to, through time-division multiplexing, sequentially address and load individual analog memory cells (47) in said matrices, wherein each memory cell comprises a storage capacitance (44) arranged to hold a weight value as an analog voltage, and wherein said control unit (9) is configured to perform periodic updates of the weight values to compensate for voltage leakage without interrupting the ongoing computation flow of the system.

Claim 7.

The system according to claim 6, characterized in that the digital control unit (9) is arranged to perform a calibration cycle during synchronized measurement windows (20, 24), comprising that the control unit (9) measures the system's current response during the differential perturbation analysis and compares it with the digital reference model (39), wherein the control unit (9) computes and applies meta-learned correction values to the memory cells (47) in the correction matrices (3, 8) via the addressing device (12, 15, 34, 46, 48) to dynamically compensate for voltage decay and hardware deviations without manipulating the external input signal (1).

Claim 8.

The system according to any of the preceding system claims, characterized in that the system comprises a discharge circuit with an impedance element arranged at said reference nodes (29), wherein said impedance element is configured to allow a higher discharge rate than the natural leakage of the memory cells (27), wherein the digital control unit (9) is arranged to, at each regeneration cycle (18), initiate a normalized reference signal (32) whose decay is accelerated by said impedance element to form a time-coded reference curve, wherein the meta-learning algorithm uses the instantaneous amplitude of the reference signal (32) as a temporal coordinate to determine an exact position in the natural discharge curve (31) of the memory cells (27).

Claim 9.

The system according to any of the preceding system claims, characterized in that the atomic triad comprises a cascade-coupled signal path wherein the first correction matrix (3) is arranged to perform a pre-conditioning of an input signal before the primary computation matrix (33), and the second correction matrix (8) is arranged to perform a correction of an output signal after the primary computation matrix (33), wherein the correction matrices (3, 8) and associated non-linear activation stages (4, 10) form a cooperating correction engine arranged to, via the meta-learning algorithm, dynamically isolate and compensate for physical non-linearities and variance in the primary computation matrix (33).

Claim 10.

The system according to any of the preceding system claims, characterized in that the digital control unit (9) is arranged to perform sampling of output signals (45, 49, 11) and differential perturbation analysis at arbitrary and selectable times by dynamically varying the temporal displacement (25) of the time window (24, 20) in relation to the regeneration cycle (18), for the purpose of extracting time-dependent gradients and calibrating the meta-learning algorithm for all dynamic states of the atomic triad (3, 8, 33) during the discharge process (31).

Claim 11.

The system according to any of the preceding system claims, characterized in that the analog matrices (3, 8, 33) constitute an atomic triad arranged on a common semiconductor substrate, wherein the first matrix (3) comprises a first cell structure (30) with interleaved computation cells (27) and reference cells (29), and that the second matrix (8) and the third matrix (33) preferably are configured to utilize the reference cells (29) in the first matrix (3) as a common reference, wherein said matrices (3, 8, 33) are arranged in immediate spatial proximity to each other in a thermally coupled layout for the purpose of minimizing differential drift within the triad through shared utilization of said reference cells (29).

Claim 12.

The system according to claim 1 or 5, characterized in that the digital control unit (9) for emulation of deviating physical states is configured to programmatically apply hidden digital offset values to at least one of matrix weights, the discharge time of the reference cells (29), a digital-to-analog converter (DAC) (2) for input signals, and analog-to-digital converters (ADC) (45, 49), wherein the control unit, during the inner loop (41) of the algorithm, fixes this digital offset pattern and varies the input signals to said DAC (2) to prevent overfitting, and that the system, upon transitioning to normal operational inference operation, is arranged to remove all emulated offset values and reset said trim-DAC (38), wherein a real-time operating inner calibration loop exclusively fine-tunes the correction matrices (3, 8), the analog output signal of which, after an activation function (10), is linearly summed with the output from the primary matrix (33) via an analog adder (35).

Claim 13.

The system according to any of the preceding claims, characterized in that the digital control unit (9) is arranged to digitally extract an analog column derivative for the second correction matrix (8) by using a search method to identify a specific row index (k) constituting a maximum carrier row which exhibits the highest instantaneous signal amplitude, wherein the control unit (9) is configured to multiply a differential oscillating voltage amplitude response, isolated for said selected row (k), by the inverse of its known static carrier amplitude, whereby the system is arranged to virtually recreate the effect of a row perturbation and calculate an intermediate sensitivity without applying additional physical transient injector circuits to the connections between the first correction matrix (3) and the second correction matrix (8).

Claim 14.

The system according to any of the preceding claims, characterized in that the system is arranged to apply said row perturbation via two separate perturbation rows, comprising a first perturbation row configured to apply a positive perturbation signal to a positive weight component ($W+$) and a second perturbation row configured to apply a corresponding inverted perturbation signal to a negative weight component ($W-$), whereby the system is arranged to generate a differential and inverted perturbation for evaluation of said differential perturbation response.

Abstract

The invention relates to a hybrid computation architecture for neural networks comprising analog matrices and a digital control unit. The architecture utilizes an inverted meta-learning algorithm configured for internal parametric adaptation, wherein the algorithm is arranged to estimate and compensate for internal hardware variations rather than external input characteristics.

Through atomic triads of interleaved analog matrices and the use of short temporal measurement windows for differential perturbation analysis, the architecture achieves a deterministic stability independent of thermal drift or component aging. The system is arranged to asymmetrically separate a primary high-frequency computation path (payload) from a low-frequency calibration cycle, enabling analog computation with digital precision at a reduced energy consumption. The architecture is scalable for both cloud-based computing systems and low-power edge computing systems.